

ActInf Textbook Group ~ Cohort 4 ~ Meeting 7

(Chapter 3 part 1)

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Duration: 55:21 | Uploaded: Unknown Transcript method: auto_caption

hello it's July 18th and we're in our first discussion of chapter three in cohort 4. so before we go to any questions or anything does anyone want to just bring up any overall thoughts they had on chapter three just about the chapter where it is in the book or what they remember or got from that chapter yeah I would add that I actually just generally think the chapter was written very well and it was very clear um I think the high road can be philosophically and mathematically very abstract so I personally appreciated the kind of clarity that the authors took to discussing things like Marko blankets and um existential imperatives I guess it's it's a the high road perhaps more than low road provokes further questions further ontological um and philosophical metaphysical questions you know the well what you know how how deep does the hierarchy of Marko blankets run or what's the fundamental base layer of this um this whole system but on the whole at least in so far as the free energy principle is a very simple explanation of something very fundamental um yeah I thought they did a really good job outlining that particular point nice thank you anyone else any thoughts low road High Road anything yeah yeah I really enjoyed um this chapter probably because there's less equations to uh get hung up on but especially section 3.7 just the idea of using active inference to reconcile um different theoretical perspectives so in my area action we have debates about whether movement commands are represented centrally versus self-organizing or are inactive this is hotly debated for for decades now um that that's something that's really attracted me to active inference is to have more unifying view um rather than keep going with the debating so I really appreciated section 3.7 for me personally any other thoughts on three or about how somebody is seeing the low road and the high road well Daniel I want to see um about the the figures when I when I look at the figures uh for example a figure 3.1 where uh uh where the figure explains the uh blanky stays um but in addition to this Blank Space there are um many other like notations and and formulas so it seems quite quite uh intimidating those foreigners like like uh for example the the first equation on the left you have X you know dot x equal to f

of x Etc so others all this notation is not explained that's you know it seems to me um that it creates uh blocks for understanding um of the uh Concepts so so Vision I thought it's it's kind of a little bit incompanied uh I know it's difficult because those are those in order to understand these Concepts you need more uh background uh explanations so so but when I read those figures I I hope to understand all the information that it contains in the figures if there are things that not understandable uh that it creates a kind of a mental block uh in my mind sure totally makes sense so and it's good to want to understand all the notation so this is a representation that is going to come up again and again um it's the Markov blanket and there's kind of two uh types of notation happening here there's like the variables that are being locally assigned and then there's some more conventional notation so like some of the conventional notation is f is a function of something and then Ω is a noise term on something that's conventional and also the dot being uh rate of change so it's kind of like having like a prime or a first derivative and then there's the locally assigned variables so external X internal μ octave U sensory y um now there is a typo in this figure a small one just it's in the erata okay but the visuals are correct each of these functions each the rates of change of each of these four kinds of states we're describing the rate of change so it's kind of like a flow map for all four of these nodes on a graph for the internal external states in the blanket right rate of change yeah and each of those rates of change is broken up into two components but kind of like a signal and a noise that's and noise could be zero or it could be all noise and then signal components on the rate of change of each of the [Music] nodes in this particular partition the signal component is a function of all the incoming arrows and itself at that time right so so in chapter two we we understand there are no road we understand the active inference as the kind of optimization problem where you can you have a choice of action and and perception so how how does actually in the perception play a role in this figure in the in the in this Mark of blanket in this uh in a figure you just showed yeah good question well we associate this incoming sensory state according uh to perception and then the outgoing statistical dependencies according to action and then what happens here which could be nothing if it's a very simple system or it could be sophisticated or unknown the same as external um we just call this the cognitive States so this is the perception cognition and action of the agents blanket plus internal States that's the particular thing uh uh insulated off from the external States so that's why this is like the minimal base graph for the cybernetic agents is having the input cognition and output and the um the work of isomera and first and others like Maps this onto neural networks that have like an input processing and output layer right okay

everything what are the choices because in the chapter two it talks about the the perception and the action are the choices made by the agent right right so those are the choices not the states and it's a decision kind of decision you can make you can choose to to to to to precept this way or the other way you can choose to take that action or or another action so here you're you're seeing the action state is is the action is the choice to um ways to think about this first is that's why we talk about beliefs about action action prior uh policy prior policy posterior so because we're talking about our beliefs about action and the second thing is this is coming from a high roads this is kind of like a forget what you knew about chapter one and two start with just this particular partition on a base graph and then the chapter continues with connecting it more to perception action but start with like this is the topology of systems that can um sense and act or you could propose a different topology of that system um young and then anyone else uh yeah uh I just wondered so the sensory states are the perception and the active states are the action intuitively and well those two are directly connected to each other and which is which means that they can bypass the cognition and it's really hard for me to imagine a state of at active States just bypassing the Coalition and just going directly to the sensory State and also vice versa so can anyone come up with a good example of it awesome question does anyone want to give a thought on that about this what is this double-headed arrow in the center so my understanding this error means that the active States is affected by by the uh sensory data at your connect so basically you can see here f_u is a function of Y why is the uh sensory data that the agent gathered from um from the external world together for me yeah and on the other hand also the U is the your action also will affect your uh sensory data y uh for example in the in the chapter two the apple and frog example if you choose your action to look at apple tree that's you if you move your eye towards Apple chain that will generate scissoring data about Apple you know what it Observer will be apple so that would be the action will affect your sensory data why and you essentially did a while I will affect your action uh yeah I'm just thinking these are all true um another thing to keep in mind is like the effect of a given Edge in a model or system of Interest it might not be relevant but it's kind of like in principle so if you have a computer system where one computer is getting input and sending it to a second one sending it to a third one and then emitting it there is no effect of this Arrow or if you have some you know some other system where this in practice is zero so just because there's an edge doesn't mean that in in that simulation or that model it's like it matters in a certain way and the reason why it's there in principle is because the definition of the markup blanket for a given node is all the nodes that kind of insulate it and the co-parents and

so just in principle they could be connected under the markup blanket uh definition like the Markov blanket nodes could all be connected to each other and be connected into the um node of Interest and also these assignments of internal external active sensory these are all relative to a given internal node of Interest from if you call this one internal then this would be sensory and that would be active or if you picked one over here it would be different so these are not like essential features of nodes these are just with respect to a chosen internal State on a created model um Darius yeah I mean um just staying on the market blanket for a bit um I've seen quite a lot of conflicting literature and opinions on to what degree it constitutes a metaphysical reality so I think one can't help but think of a kind of cell when you think of a Markov blanket and the blanket states that are the boundary between the cell and the external environment to a degree or Markov blankets from the biggest markup blanket a universe with the universe all the way down to the smallest of Markov blankets and all the ones embedded in between to what degree do current um free energy principle Ferris believe that there's a genuine metaphysical reality or is it just a way of visualizing and understanding statistical regularities yeah big question what does anyone think budget or anyone else um I guess like I would think it would uh so would that be connected to like our subjective experience would be this statistical model so that would be the Markov blanket is that what you mean or I mean I don't mean that it's Observer uh relative you have the universe as is and our observations indicate to us that there are um separate patterns of existence things that are distinct from one another uh-huh all those boundaries which are defined by the markup blanket metaphysical as if they exist I mean I think the only way of kind of conceptualizing that is as a material distinction or do they just exist do they solely serve to be a statistical representation of the actual distinctions that we observe but perhaps there is no physical substrate underlying those regularities that's my question yeah it's a huge question there's there's a ton to say in it definitely a lot of perspective so like there's a way broader um Range one topic that comes up in that setting is the different flavors of instrumentalism and realism and scientific realism pragmatism so instrumentalism is like the least metaphysically committed and just says look linear regressions help us understand the relationship between this and that that gives a sort of cognitive realism to the linear regression it might influence someone's life so it does matter so it doesn't mean that we don't take it seriously but no one is going to mistake the map for the territory um then for various uh reasons there's uh also more of a realist not necessarily like a platonic realist but like a scientific realism it's like yeah the objects that are under study are real now the finger isn't the moon so pointing at the phenomena it's still like not map territory

the model and the equations is not a system that's obvious so just saying but the model doesn't have this feature of the system is like yes but we're talking about the reality of this system you know six people in the elephant they can still be talking about the reality of the elephants that's kind of a scientific realism and uh Mel andrews's work and the math is not the territory live stream 14 and a lot of other discussions on that and then um there's a lot more to say but that's a great question area friend I was when I read this chapter I I met uh the example that I thought oh well I intuitively kind of found helpful was playing the piano and I wondered if you could verify uh the veracity of that example if it doesn't indeed explain many of the Dynamics between my internal States my active States the hidden States and in turn the sensory states that I received to complete an action uh kind of feedback loop and I think it also might explain the uh um The Edge is between um the active States uh in the example of playing the piano and uh the sensory state that might ensue should you take a particular action all in this instance as I hope to explain not take a particular action so I I thought of the piano kind of sitting at the piano with the intention of playing and I the the I believe I want to play piano I'm not 100 certain of what might be fall under you know what can um would fall under the internal States but other than a belief that I can play I'm sure there are many many other factors there that would be um necessary for one to um play initiate the action but then I touch a key a a note and that is that has that entrains an effect on the perception of the piano which then in turn I don't I don't play that note directly um I elicit sound from that from the piano which then in turn uh gives me the noise which hopefully sounds good and so I um continue that that effect by playing the next notes in accordance with the previous one and so I build the tune as I go hopefully and hopefully that's pleasing rather than full of kind of dissonance and offbeats and and irregularities and I thought about the relationship between the active States and the well and the and the the blanket States the sensory States if you were to Omit a note they buy inhibiting the effect of the action on the piano thereby if you won't perceive anything in its you won't get anything in return so to speak from the piano because you have emitted a note so yeah I kind of thought about this around playing the piano and yeah awesome that's a really cool example and um in chapter seven there's gonna be like a musical example and music cognition there's a lot more work on as well but in this example like the wrong note is played quote wrong note different than the expected tune but the expected the expected note is still heard because the prior is so strong and so like um Ali and and several others have have a lot of interest in in that um music perception it's like very rich with with music perception and generation improvisation it's a really cool area Daniel if I can just ask if you can go back to

diagram 3.1 I think it was I just wanted to clarify something that's just come to mind now and I'm not very good at all the uh all the technical details yet but how is a uh the noise from the external States how does that how does that relate to how does a correlate correlate to the impact or the effect of the noise at the sensory states is it a how is that kind of transduced I'm thinking of the piano example again um how does that dissipate through the is it equivalent so a particular um noise at the external state is it how is that experienced or understood as how is it similar or dissimilar from the the noise perceived at the um or generated in the sensory state very interesting question so think about each of these four nodes as being like a sensor or a thermometer um in in and of themselves like their probes that are tracking these flows so what would it mean to have um this be high and this below well maybe the um temperature in the ecosystem is constant but it has a lot of variability so there's like no signal really but there's a high variability but we have a very accurate thermometer that we're taking in information from or conversely maybe it's a deterministic temperature outside and our thermometer is highly variable so this is kind of just like the outline that gets filled in like a hundred percent during the modeling process but this is just extremely General outlining like where the city just saying that each of them are partitioned into a signal and noise component in principle but but that's why it really helps to also to think about specific examples um uh or go ahead oh yeah um like because I'm a psychiatrist uh uh what what keeps in my to my mind about this micro blanket is that it's really a cool concept you think of the process of psychoanalysis because like the therapists by using by using his own internal State and inferring the internal state of the patients added by making decisions and influencing these sensory States and these active States active States active States active States activities to change the internal State or the model internal model of the patient so I mean that's quite interesting because like the basic concept of Markov blanket is that you can't really have access to external States which means that you don't have any access to the internal state of others but you can change the internal state by having very good inference about it and doing the right active actions or providing good sensory States like make maybe making the patient comfortable or sometimes you you use um many different techniques with Psychotherapy such as hypnotics or I don't know maybe other relaxation techniques so well the basic concept Markov blanket is really cool for me so like it wasn't a question it was just Electronics that's awesome thank you that is absolutely the core of the Markov blanket concept definitely a lot of philosophical like smoke and fire is in that area for for I think a variety of interesting reasons partially because it touches on like how things really are in some people's eyes or it's something that seems to

explain like everything and nothing but at the end of the day it's just a description of the sensorium and the real important um missing the the missing piece or the missing Edge not missing piece that really embodies the constraints here is is the no telepathy no telekinesis clause no Edge between internal and external States except via insulating nodes so by making a fully insulating shell then you you don't need to make any claims about what's in the on the other side of the blanket now you could you might happen to know or you might have a really good model of what is happening or something like that but those models aren't free information takes energy um and they're always understood as projections of of onto hidden States or trackings of hidden States so it's really just a recognition of the sensorium and the cybernetic system boundaries with actually a lot of generalizations that make it really exciting and even more flexible than people might initially consider just from looking at this relatively simple system Darius yeah I was um I was wondering this seems for this makes perfect sense to me in terms of a sentient uh non-steady State equilibrium system like a human brain um but if in terms of this equation in terms of the markup blanket generally but to what degree can we think that entities that don't act that exist in steady state equilibria so you know like for instance classic example of a drop of oil in water can we I understand that you can't necessarily attribute action to the drop of oil um or at least purposeful action when we look at that diagram in 3.1 do we have to think of it as if the drop of oil it's as if they're behaving in this manner or are we committed to some notion of course the external and the internal state but they're maximizing its model evidence it's actually actually is sensing and it is being active so where where is the kind of dividing line here between sort of sentience and and non-sentience and to what degree if there isn't if there is one is it just that the drop of oil basically exists it just seems to be maximizing its model evidence but there's no real tell us to it yeah you know that because the action here the active inference seems to me to be very highly linked to sentience great great questions well one angle is an instrumentalist goes into that area without feeling like they're taking on um uh descriptions like it's a sentient as they extract the signal of so that's very like relational or sort of like perspectival view on sentience um where somebody taking on more ontological commitments would say that they are um exploring or characterizing the true nature of sentience or intelligence here's a really early 2006 or so image from a friston paper this is the winged Snowflake and it's like systems that persist conditioned on seeing this snowflake it's not conditioned upon all possible you know a room that's that's 100 degrees C conditioned on seeing this snowflake there's going to be snowflakes that through wind blowing them around so just doing no better than you'd expect from the wind

currents or doing better than the wind currents they preclude phase transition of melting once they cease to exist then they cease to exist so that's like one um because you asked about how to describe these simple systems too and then this path in a girl's paper made the clearest taxonomy of systems from inert systems where like some of these nodes are trivial so some of these nodes can be trivial or null in effect and that's what gives the ability to Model A really simple system but then the composability of the Bayesian graph um format allows the the construction of also more sophisticated cognitive models all keeping the markup blanket condition because you could have internal States could just be like a pass-through or you could like imagine they functionally didn't exist or it could just it could be this or it could be that um Andrew I just wanted to add to that that um I mean in the image after itself later on I guess section 3.6 page 56 um they attempt to kind of provide their own answer to the distinction between like a sentient or perhaps non-sentient um organism and they just say this is one answer so not you know an exhaustive answer to the question but um like having the capacity for like for an organism to have the capacity to model alternative Futures and to have that level of temporal depth um that that's one answer that makes sense to me in thinking about like the drop of oil on the water like Perhaps it is you know it's reacting to something like sensory input and acting but nonetheless like I can't imagine that the the oil is you know anticipating future um you know sensory information and and coming up with its own representation of the proper action to take in the future and what it should be moving towards so that's kind of how I've viewed you know a way of deriving an answer for the sentience versus non-sentients question cool thought and yeah like as an inner object or even a simple active object may not provide any evidence like you don't need to appeal to any counterfactual generation in order to explain even potentially life-like behavior and certain you know basal cognitive behavior may not need explicit entertaining of alternatives uh wanchon yes um I have the question about the figures about the direction of the arrows can you go back to the figures 3.1 right so um if you look here there are um on the upper right there is uh two uh Direction figure between active States and internal States what I'm thinking is that the internal will affect the active stage right so what action you take depends on your internal state but how does active State actually change the internal stance directly yeah yeah totally amazing the internal States will be affected by your sensory uh sensory stays right this isn't what are the sensors this isn't what does happen in a given model this is what can happen in one arrangement of an abstract-based graph so don't connect it to the actual structure of causality even in the world this is just saying this is like a template sketch of a cybernetic agent with

and what's mainly ruled out are the telepathy and telekinesis and the other one that you can think of as ruled out instead of asking why this is two-edged in principle what's ruled out backwards causation of action we wanted the entirely in control of action and backwards causation of sense States we don't want to have our thumb on the scale so there's just many many ways to think about it and in any actual modeling situation this level of abstraction is not engaged with it becomes a lot clearer when you're talking about specific functions describing different states in a specific system of interest uh young Hood oh yeah it's a totally different question but like considering the reinforcement learning sometimes when you sample a lot of sequences and then like like there are some models like Tomlin eichenbaum machine or success representation which actually updates the state to state transition based on the experience that you had um it it even though you don't really perceive additional information you can update the transition metrics based on your memory so you replay and Playback the memory and it actually updates your um internal States so considering that fact um where should the where should we put the memory Replay in that figure is it like do we have to say that the memory replay is some kind of um action or is there another alternative solution great question I'll just try to um set it up indirectly because I think it's a it's a it's a real um direction is there's going to be kind of two ways Main families of ways of representing the cybernetic agent that we're going to see one of them is this Markov blanket type so here it's like a very physics base that's why we're coming from the high road with the free energy principle because it starts from this kind of like flow of physics space the other representation which we saw in chapter two and gets used in chapter four and so on is this um Bayesian generative model but like you don't see the um internal external and blanket States though here's different notation but they're both um ostensibly about the same active inference agents right where one can view the observations here as the sensory States and then one can understand that the actions taken have causal impact on the world but that's reflected by the B Matrix and learning on the B Matrix learning how actions are associated with changes in the world when it's coming more from like a high road perspective that's more in the physics and that's where you see more of the physics type formalisms and then in more of the um uh decision making and Signal processing area from the low road that's where we see more of these like not physics based in principle Bayesian methods there's probably a lot more to say about that and other way to say it but um like this representation is about a cybernetic agent also this representation is about a cybernetic agents and they meet inactive inference where those models have some relationship figure three points or does anyone else want to make a comment or question I

wanted to uh just follow up on that note we look at equation 3.2 a very similar to equation 2.5 except then 3.2 operational free energies contingent on a model so I'm trying to relate that with the previous comment about the cybernetic unit in the Markov blanket is that related to the model the contingent model yeah exactly so free energy like expected free energy variational free energy it's a quantity about a model like sum of squares is a quantity about a given linear regression that you might do as like an error term if you're fitting a regression variational free energy is like an error term that is about a specific model I mean if you put another node you made a different model the free energy would be a different number um if you didn't have a model a map then you can't calculate a free energy for a model that doesn't exist so it's a descriptor of a map so everything is really conditioned upon the parameters and the values that are in that specific model that is yeah Darius yeah um just maybe a point of clarification for myself I presume it can't be entirely internalist um because at least just like looking at the KL Divergence in the equation the difference between the approximate posterior and the true posterior requires some commitment to their being an external state that you're trying to predict so where is is that just you've got the variational model uh so you've got the variational for energy which is tied to your model but ultimately it's also on you know uh subserved by the fact that you are an agent in the world and that you're making Bayesian inferences about the world and it can be closer to the real world state or not I just don't know if I've got that the right way around yes that's true and the Kullback-Leibler Divergence gives us this optimizable incremental way to update our beliefs cue to bring them to um where where they aren't moving anymore just as a first approximation where we we can't believe any better about perception or action or cognition with respect to as you pointed out the the the underlying true distribution but this can be understood in practice as just not wanting to update your belief achieving preferred States which is the maximum evidence model for the incoming data if you literally knew where what photons were going to come at you next and that is um minimized surprise so this is kind of just like this three parts are they're they're being equated here but they're different framings this is saying this is the most performant model best model evidence this is the least surprising model if you knew how much to be surprised you would have the best performing model this is a well understood relationship this is saying because of how free energy is defined which isn't invented by active inference lookup energy-based method or evidence lower bound this quantity is tractably optimizable and if we can reduce the Divergence um where there's the KL diversions shrinking will be doing strictly better that doesn't mean good enough necessarily but strictly better towards situation yes um I have the question about this

model concept does the model refers to others Dynamic equations in figure 3.1 what what do you mean by model here here yeah yeah the model we have about how y are generated so so it could renew everything in in figure 3.1 including like the the because Y is generated based on your internal State based on external state right also it's also affected by the active State yeah if you had the total understanding of what photons would come next including how the world changes and how your action influences it that's the this is like an agent level loss function for understanding how y are generated but you could use energy-based methods on any Bayesian distribution but this is just introducing an important relationship amongst surprise minimization equals model evidence maximization that's the first part and variational free energy is a well-used method to attractively optimize this interface right so so 3.2 QX is a decision variable so to minimize you mean or to maximize you want you want to minimize it right so is the total beliefs those are the beliefs that we control but take it again from from just this level of the high road so not explicit counterfactuals yet just flow across these four states so the the generation of Y actually are connected in all those four equations or four states right how those four states are changed uh how this works this changes so M refers to all the um all the this process how this process are described yeah M could be a linear regression and Y would be the data and then we'd say the surprise is minimized the evidence is maximized and the variational free energy is minimized when we have the model and the data and the models as good as it can be for the data so it's the exact same thing like the two pieces that come into play here are the model and the data right so so the decision is your QX all right so it's a function it's not just the decision though QX refers to all of the optimizable beliefs not just about action all right so I mean I mean it's it's a that is a it's a belief that you you may choose to minimize your total surprise right or to minimize your variation of so among all you know QX you you you choose one to minimize this expression on the right side of 3.2 is that right so Qx what is X here x does it refer to the um internal state of their mind X is the external States those are the hidden causes about how y figure um 3.1 X external States how sensory causes are generated no thumb on the only the external World causes sensory um degenerative model of sensory States being produced as they are maximize your evidence about incoming data minimize your surprise by incoming data minimize your variational free energy about incoming data all of those are causal modeling of external States analogously to how internal states are like doing that for action because which one you call internal and external it it you could flip into the other one at this level of generality not when you actually build a model but this is there's still nothing that's really fundamentally different between internal um and

external other than which side the modeler chose to do internal Darius this will be my final question I promise um I just wanted to ask about the kind of architecture of the generative model so the agent is constituted by the internal States and the Markov blanket and the agent this was in Ben White's talk as well there's this point of potential conflict or confusion that the agent has a generative model or the agent is a generative model and that General generative model has priors based on its optimism bias of what it what of the preferred states to attractive States it wants to be in is it that can we consider the generative model therefore the internal states which are encoding beliefs about the world and the Markov Blanca and in that case how how could you say if it is that why would you say that the agent has a genitive model and if you do hold that if you hold that claim what's extra about the agent beyond the genotype model yeah those are great questions well what's extra about a person beyond their health record which you do a linear regression on I mean the territory itself so that's the instrumentalist answer is this is all just about the model we're just talking about base graphs and statistics so we're obviously talking about maps not territories so that's again trying to take on minimal commitments but the question about like does it have a generative model or is it a generative model well if you're talking about a software agent you designed they might be the same thing you might say it literally does active inference because I I actually have it implementing that here's the script but if it isn't literally slash actually an active inference simulation it seems like you're going to be in the situation described in figure 9.1 so you're here you're building this is the figure 4.3 generative model for the subject of a behavioral experiment that's why it's subjective it's their perspective as the Mouse um in the in the Maze or whatever but it's also like they are subject to the behavioral experiment and then here's the Outer Loop so would you say that the mouse is the cognitive model that the researcher made probably not so in practice a lot of this like metaphysics doesn't come into play when you're talking about a specific generative model it's a lot clearer that it's just like a very hopefully well well uh designed but just a reduced model of the system of Interest that chapter nine is about like this kind of empirical modeling ninky oh I cannot hear you actually can you hear me now yeah yeah go for it okay great um no I have a question that's very much related to that I think there are lots of sort of General comments about the systems to which we can apply active inference um and what sort of the specific form of criteria are to describe a system as adaptive um and I'm also very much interested in these philosophical perspectives and seeing sort of how um in which sense you can you could view a whole organism um as um as sort of following the principles that specif specified by active inference or in which sense you could

sort of use active inference to describe maybe systems at our lower lowering within that hierarchy that are maybe like um smaller regulatory systems in the body um so I was just wondering if there's any um yeah sort of research on uh sort of the general applicability of active inference as sort of a way to analyze adaptive systems um and I think a lot comes down to the mathematics which I don't fully understand yet but or awesome question that's very cool One Direction of research that comes to mind is uh like here's a 2019 paper so there's been a lot since then and a lot more integration with active inference too but um some of these representations of um multi-scale systems nested systems so being able to um in like chapter five in the textbook there's going to be like a motor circuit that's influenced by a higher order decision-making circuit so then you you maybe you may be able to to sketch out some of those lower level units which paper was this again I'll put it into the um I'll put it into the chat thank you all right well cool um about chapter three I guess if uh next week or in in yeah next week at the other time we'll come back and if people can have added or voted on different questions and we'll talk more about the questions next week so thank you see you all thank you Daniel thank you